

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

2 Artificial Intelligence (sbh11)

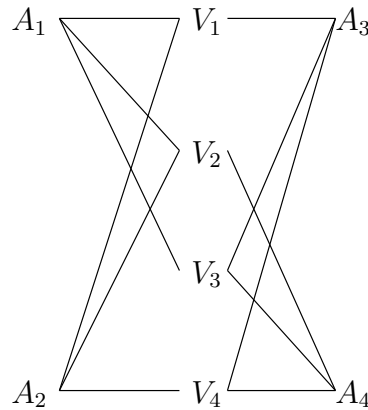
A *constraint satisfaction problem (CSP)* has four variables V_1, V_2, V_3, V_4 , each with domain $\{1, 2\}$. The constraints for the problem require that given any three variables exactly one must have the value 1.

Answer: Throughout the solution notes, numbers in brackets provide a *very approximate guide* to allocation of marks.

Representing a CSP using only binary constraints.

- (a) Explain how this problem can be represented as a CSP that uses only binary constraints. Illustrate your answer by giving a graph representing the problem. [4 marks]

Answer: The constraints for the problem require that, for example, taking variables $\{V_1, V_2, V_3\}$ they can take assignments 122, 212 or 221. Similarly for the other three subsets of three variables. To make the constraints binary we introduce new variables. So introduce variable A_1 associated with $\{V_1, V_2, V_3\}$ and having domain $\{1, 2, 3\}$ (1) where each element of the domain corresponds to one of the possible assignments to the three original variables (1). The binary constraints are then $\{A_1 = 1, V_1 = 1\}$, $\{A_1 = 1, V_2 = 2\}$, $\{A_1 = 1, V_3 = 2\}$, $\{A_1 = 2, V_1 = 2\}$, $\{A_1 = 2, V_2 = 1\}$, $\{A_1 = 2, V_3 = 2\}$, $\{A_1 = 3, V_1 = 2\}$, $\{A_1 = 3, V_2 = 2\}$, $\{A_1 = 3, V_3 = 1\}$ (1). New variables A_2, A_3 and A_4 are associated in the same way with the remaining three subsets of the original variables. As a graph (1):



Forward checking in a CSP

- (b) Describe how *forward checking* can be used to aid the search for a solution to a CSP. Illustrate your answer by showing how it applies to the problem in Part (a), for assignments $V_1 = 1$ followed by $V_2 = 2$. [4 marks]

Answer: Each time we make an assignment, say to V , forward checking involves removing from the domains of each variable involved in a constraint with V , all remaining values whose use would invalidate the constraint (1). For the given examples:

Variable	V_1	V_2	V_3	V_4	A_1	A_2	A_3	A_4
Domain	12	12	12	12	123	123	123	123
$V_1 = 1$	= 1	12	12	12	1	1	1	123
$V_2 = 2$	= 1	= 2	12	12	1	1	1	23

(1 - A_1, A_2, A_3 become 1; 1 - correct domains for A_4 ; 1 - other V_i unchanged.)

Arc consistency
and the AC-3
algorithm

- (c) Describe the *AC-3* algorithm for imposing consistency in a CSP. Include in your answer descriptions of an *arc*, what it means for an arc to be *consistent*, how a non-consistent arc can be made consistent, and the overall operation of the algorithm. [6 marks]

Answer: If V and V' are in a constraint there are two corresponding arcs $V \rightarrow V'$ and $V' \rightarrow V$ (1). An arc $V \rightarrow V'$ is consistent if, for each possible assignment to V (1), there exists an assignment to V' that respects the constraint (1). An inconsistent arc is made consistent by deleting any assignment to V for which there is no suitable assignment to V' (1). The AC-3 algorithm works by employing this method and, each time it fixes an arc $V \rightarrow V'$, dealing with the fact that this can make any arc $V'' \rightarrow V$ inconsistent.

```
NewDomains AC-3 (problem) {
    Queue toCheck = all arcs i->j;
    while (toCheck is not empty) {
        i->j = next(toCheck);
        if (removeInconsistencies(Di,Dj)) {
            for (each k that is a neighbour of i)
                add k->i to toCheck;
        }
    }
}

Bool removeInconsistencies (domain1, domain2) {
    Bool result = false;
    for (each d in domain1) {
        if (no d' in domain2 valid with d) {
            remove d from domain1;
            result = true;
        }
    }
    return result;
}
```

(2 - description of algorithm.)

Constraint
propagation

- (d) Consider again the problem in Part (a). We initially have no assignments, and start by setting $V_1 = 1$. Explain in detail what happens if we attempt to adjust the domains to impose consistency. [6 marks]

Answer: By setting $V_1 = 1$ we make arcs $A_1 \rightarrow V_1$, $A_2 \rightarrow V_1$ and $A_3 \rightarrow V_1$ inconsistent and to fix this we need to change the domains of A_1 , A_2 and A_3 to $\{1\}$ (1).

A_1 has arcs coming from V_1 , V_2 , V_3 . Taking the first of these, $V_1 \rightarrow A_1$ is consistent. For $V_2 \rightarrow A_1$, if $V_2 = 1$ we require $A_1 = 2$, which is not possible; if $V_2 = 2$ we can assign $A_1 = 1$, so this is OK. The same argument applies to V_3 , so we need to restrict the domains of V_2 and V_3 to 2 (2).

A_2 has arcs coming from V_1 , V_2 , V_4 . Applying the same reasoning forces us also to set $V_4 = 2$ (1).

Having modified the domains of V_2 , V_3 , V_4 we will now have to consider the arcs from A_4 to these variables. This will result in all three possible values for A_4 being deleted as each requires one of the corresponding variables to be 1. At this point we have detected that no solution can be obtained (2).
